Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - IV

Course Code: BCA431 **Max Marks: 100**

Course Name: GRAPH THEORY **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Enhance research, inquiry and analytical thinking abilities.

CO2: To improve proof-writing skills.

CO3: Apply the basics of Graph Theory to solving practical problems.

Section A **3*10=30**

1)	Determine whether the given graph is Eulerian, Hamiltonian, or neither. Demonstrate your answer.	**3**	(CO1, L3)
2)	Discuss what a spanning tree is and provide one example of a spanning tree for the graph having vertices {1, 2, 3, 4} with edges {(1, 2), (1, 3), (2, 4)}.	**3**	(CO2, L2)
3) a)	Analyze and explain why a full ary tree with 15 vertices has a different number of leaves compared to a full ternary tree with the same number of vertices.	**3**	(CO3, L4)
	OR		
b)	Examine a simple graph with 5 vertices and analyze how many edges it can have without forming any cycles (i.e., determine if such a graph is possible).	**3**	(CO3, L4)
4)	Define a spanning tree and construct one for the graph shown below, which consists of vertices {A, B, C, D} and edges {(A,B), (B,C), (C,D), (D,A), (A,C)}.	**3**	(CO1, L3)
5)	For a simple graph with 5 vertices, determine the maximum number of edges it can have and explain your reasoning.	**3**	(CO2, L2)
6)	Determine the number of edges in a simple graph with 8 vertices if every vertex has a degree of 3.	**3**	(CO3, L3)
7)	Examine whether a simple graph with 6 vertices can have exactly 7 edges, and justify your answer.	**3**	(CO1, L4)
8) a)	Examine and differentiate between a tree and a spanning tree in graph theory. Provide one example for each to illustrate your explanation.	**3**	(CO2, L4)
	OR		
b)	Analyze whether a simple graph with 6 vertices can have exactly 7 edges, and justify your answer.	**3**	(CO2, L4)
9)	Describe the structure of a full ary tree and explain how it differs from a regular ary tree in terms of its leaf nodes.	**3**	(CO3, L2)

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10)	Determine whether a given graph is Eulerian, Hamiltonian, or neither. Construct an example of each type and briefly explain why your examples fit their respective categories.	**3**	(CO1, L3)

Section B **7*3=21**

11)	Consider a simple graph with 6 vertices and 9 edges. Determine (i) whether such a graph can have exactly three vertices of degree 3, or not; (ii) if possible, construct an example of such a graph; (iii) prove why the construction in part (ii), if it exists, must adhere to certain properties.	**7**	(CO2, L3)
12) a)	Discuss the properties of an ary tree and explain how these properties are reflected in a full ary tree, using specific examples to illustrate your points. Additionally, describe the relationship between the number of vertices and edges in any tree, and prove that every tree with vertices has exactly −1 edges.	**7**	(CO3, L2)
	OR		
b)	Discuss the concept of a spanning tree in graph theory by (i) defining it and (ii) illustrating how to find a minimum weight spanning tree for a given weighted graph using Kruskal's algorithm.	**7**	(CO3, L2)
13)	Discuss the concept of graph complement and determine the number of edges in a simple graph with 8 vertices where each vertex has an odd degree. Justify your answer.	**7**	(CO1, L2)

Section C **7*4=28**

14)	Discuss the concept of edge number in simple graphs and prove that any simple graph with an even number of vertices must have at least one pair of non-adjacent vertices. (i) Elaborate on the significance of this property in relation to graph connectivity. (ii) Illustrate your proof using a specific example of a simple graph.	**7**	(CO2, L2)
15)	Discuss the differences between Eulerian and Hamiltonian graphs with examples. Additionally, illustrate how Leonhard Euler's solution to the Konigsberg bridge problem laid the foundation for Graph Theory.	**7**	(CO3, L2)
16) a)	Construct an Eulerian graph with at least 6 vertices that is not Hamiltonian. Demonstrate your solution by drawing the graph and explaining why it satisfies the conditions for being both Eulerian and non-Hamiltonian. Additionally, explain the significance of the Konigsberg bridge problem in the context of graph theory.	**7**	(CO1, L3)
	OR		
b)	Determine the number of edges in a simple graph with 8 vertices where each vertex has an odd degree. Additionally, construct one such possible graph and represent it using its adjacency matrix.	**7**	(CO1, L3)
17)	Discuss the concept of graph isomorphism and explain how to determine if two simple graphs are isomorphic by considering their degrees and adjacency properties. Provide an example of two simple graphs with at least five vertices each, and justify whether they are isomorphic or not.	**7**	(CO2, L2)

Section D **7*3=21**

18)	Examine a simple graph with 6 vertices and analyze its properties based on the given adjacency matrix. (i) Construct the graph from the matrix. (ii) Determine if the graph is bipartite, providing reasons for your answer. (iii) Calculate the number of edges in the graph.	**7**	(CO3, L4)

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19) a)	Construct a 4-ary tree with 15 vertices such that it has exactly 5 leaves. Determine the number of internal vertices in this tree and explain how you arrived at your solution.	**7**	(CO1, L3)
	OR		
b)	Demonstrate the construction of a spanning tree for the given graph G, where G is a connected undirected graph with 8 vertices and 10 edges. Show that your constructed spanning tree satisfies the properties of a tree (i.e., it is acyclic and connected). Furthermore, explain why any additional edge in G cannot be included in this spanning tree without violating its definition.	**7**	(CO1, L3)
20)	Discuss the concept of a simple graph and its properties, specifically focusing on the relationship between the number of vertices and edges. (i) Prove that in any simple graph with an even number of vertices, there must exist at least one vertex with an odd degree. (ii) Given a simple graph G with 7 vertices and 9 edges, explain how you would determine if it is possible for all vertices to have the same degree.	**7**	(CO2, L2)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

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